Weekly Research Journal

Centerline Debugging and Map Coordinate Alignment

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Abstract

This week recovered the centerline-generation and localization pipeline, identified why the previous real-car centerline did not align with the vehicle pose, and validated a repeatable map-saving procedure. A NumPy 2.2.6 and OpenCV ABI conflict was bypassed without modifying the global environment, after which the generator produced 500 valid waypoints. The VESC driver was found commented out of the active launch description; restoring it returned IMU publication to approximately 49.99 Hz, preserved LiDAR at approximately 9.97 Hz, and restored Cartographer map-to-base_link. Pure Pursuit then ran, but selected a target outside the live map and saturated steering because the old centerline and live localization used different map origins. A same-session PBStream-to-map workflow was validated. Closed-track remapping and supervised real-car Pure Pursuit validation remain pending.

1 Weekly contribution

The investigation was staged from the first failing component toward downstream control. Motor actuation remained disabled and the Ackermann command was remapped to a debug topic. Table 1 distinguishes recovered software functions from the remaining closed-track test.

Table 1: Evidence and scope of the 6 November results.

Contribution	Evidence	Scope
Centerline generator	Compatibility workaround; 500 points and 501 CSV lines	Completed
Localization chain	IMU 49.99 Hz, LiDAR 9.97 Hz, map-to-base_link restored	Completed
Root-cause analysis	Old centerline outside live-map coordinate range	Completed
Final vehicle validation	Same-session closed-track products not yet collected	Pending

2 Debugging strategy and centerline recovery

The generator, waypoint file, Pure Pursuit input, Cartographer transform, and sensor inputs were checked in that order. This isolated each blocking condition before interpreting downstream behavior.

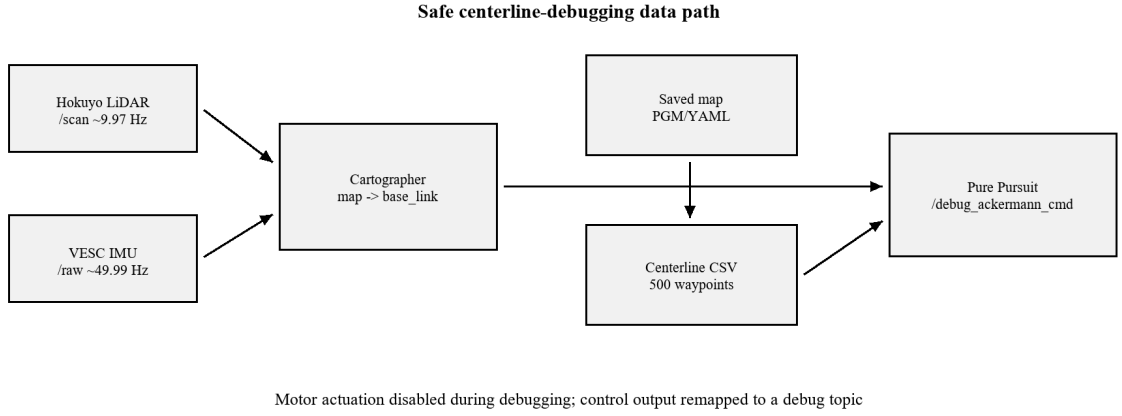


Figure 1: Safe debugging path. Pure Pursuit used the restored vehicle pose and the saved-map centerline, but its output was redirected to /debug_ackermann_cmd.

2.1 NumPy and OpenCV compatibility

The centerline node failed at import cv2 because user-site NumPy 2.2.6 was incompatible with the ABI used by the system OpenCV installation. The errors were _ARRAY_API not found and numpy.core.multiarray failed to import. Rather than downgrading NumPy globally and risking the Open3D/ZED environment, user-site packages were disabled only for this command. The system pair, NumPy 1.21.5 and OpenCV 4.5.4, imported successfully.

Table 2: Centerline-generator recovery evidence.

Check	Observed result	Conclusion
Default Python environment	NumPy 2.2.6; OpenCV import failure	ABI mismatch
PYTHONNOUSERSITE=1	NumPy 1.21.5; OpenCV 4.5.4	Compatible system pair
Generator output	500 numeric waypoints; 501 CSV lines	Generation completed
Pure Pursuit load	Loaded all 500 waypoints	CSV path and format valid

3 Restoring sensors and localization

3.1 Missing IMU publisher

Cartographer subscribed to /sensors/imu/raw, but the topic had zero publishers even though the VESC device was present at /dev/ttyACM1. Inspection showed that the VESC node was defined in robot.launch.py but commented out of the returned LaunchDescription. Enabling the node and rebuilding my_robot_bringup restored the publisher.

Table 3: Localization-chain recovery measurements.

Stage	Evidence	Interpretation
Before launch fix	IMU publisher count 0; Cartographer subscription count 1	Input absent
After launch fix	VESC firmware 6.6 connected	Driver operating
IMU stream	approximately 49.99 Hz	Recovered
LiDAR stream	approximately 9.97 Hz	Stable
Cartographer pose	Translation [0.095, -0.002, -0.115]	map-to-base_link available
Cartographer RPY	[0.972, -2.613, -0.194] degrees	Orientation available

3.2 Pure Pursuit entered the control loop

With both sensors available, Cartographer created a valid trajectory and Pure Pursuit entered its normal loop. This confirmed that the chain from sensors to vehicle pose to waypoint processing was operational. The resulting command was still withheld from the vehicle because its target selection was physically implausible.

Table 4: Safe-debug Pure Pursuit snapshot.

Quantity	Observed value	Immediate interpretation
Vehicle pose	(0.09, -0.01), yaw 0.00	Near live-map origin
Target	index 15 at (3.13, 8.71)	Unrealistically distant
Steering	0.450	Saturated at configured limit
Speed command	0.70	Debug output only; not actuated

4 Map-coordinate mismatch

The live Cartographer map covered approximately x = -1.75 to 3.85 m and y = -4.55 to 4.75 m. The old centerline began near (2.95, 9.01), outside this live-map range. Thus, two datasets named map were not geometrically interchangeable: their coordinate origins came from different mapping sessions.

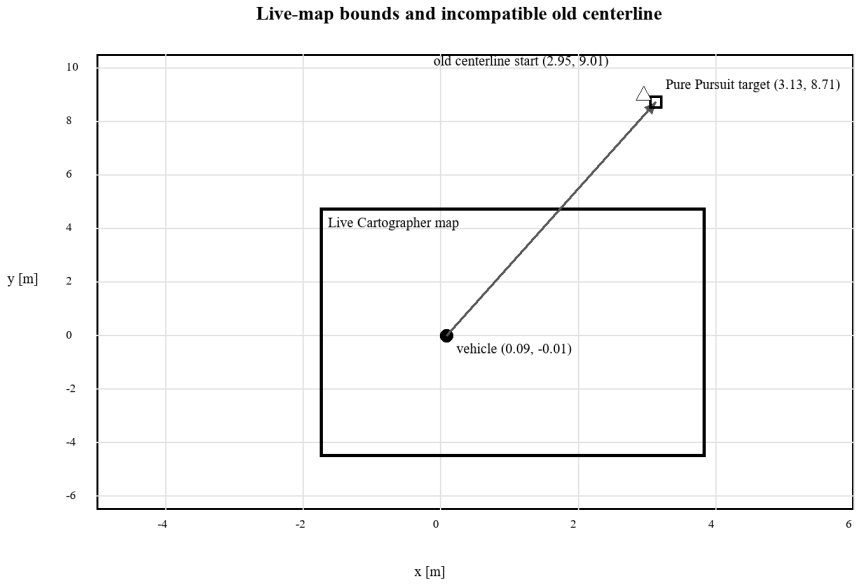


Figure 2: Coordinate evidence for the Pure Pursuit failure. The vehicle lies inside the live map, while the old centerline start and selected target lie beyond its y-extent. This explains the distant target and saturated steering.

Table 5: Map and PBStream cross-check.

Source	Origin [m]	Image size	Assessment
22 Oct my_map.pgm	(-3.73, -4.84)	317 x 433	Original centerline map
my_map.pbstream conversion	(-4.40, -6.03)	188 x 194	Does not reproduce map
map.pbstream conversion	(-2.35, -2.10)	145 x 157	Does not reproduce map
New laboratory map	(-2.05, -4.55)	119 x 195	Valid save; not a closed track

5 Validated same-session map workflow

The active trajectory was finished, written to track_20261105.pbstream, and converted from that exact state into PGM/YAML at 0.05 m resolution. This validates the persistence procedure needed to keep localization and planning in one coordinate system.

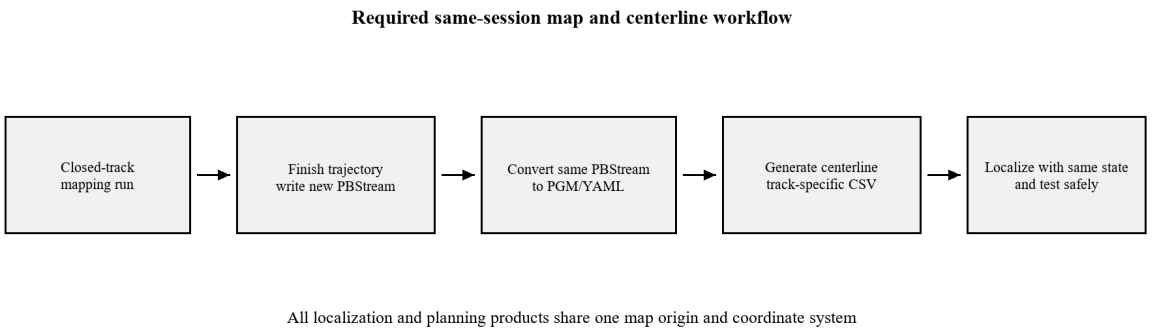


Figure 3: Required workflow for the next closed-track test. Reusing a map or waypoint file from another mapping session is excluded by construction.

Table 6: Reproducibility record for the validated save sequence.

Operation	Artifact or setting	Status
Finish trajectory	trajectory_id = 0	Validated
Write state	track_20261105.pbstream	Validated
Convert state	PGM/YAML from the same PBStream	Validated
Map resolution	0.05 m	Recorded
Laboratory centerline	Extraction failed because environment was not a closed loop	Expected limitation

5.1 Interpretation

The laboratory centerline extraction failure is not a new software defect: the algorithm expects a closed race-track geometry. The save-and-convert chain is nevertheless useful evidence because it confirms that a future track-specific PBStream, map, and centerline can be generated from one session.

6 Completed work and open work

Table 7: Project status at the end of the reporting week.

Status	Item
Completed	Reproduced and bypassed the NumPy/OpenCV compatibility failure.
Completed	Generated and loaded 500 centerline waypoints from the old map.
Completed	Restored VESC IMU, LiDAR, and Cartographer map-to-base_link.
Completed	Identified map-origin mismatch as the centerline-alignment cause.
Completed	Validated finish_trajectory, write_state, and PBStream conversion.
Pending	Create a same-session closed-track PBStream, PGM/YAML, and centerline.
Pending	Localize using the matching saved state.
Pending	Verify nearby targets and non-saturated steering in debug mode.
Pending	Perform supervised real-car Pure Pursuit validation.

7 Discussion and next experiment

The main software failures are resolved, but the old centerline is not valid for the live localization state. The important result is therefore both corrective and procedural: map origin is part of the experimental state, and matching filenames or frame names do not prove coordinate compatibility.

The next experiment should perform one closed-track mapping run, save the PBStream and PGM/YAML from that session, generate a track-specific centerline, and localize against the same state. Pure Pursuit should first remain on /debug_ackermann_cmd until target points stay near the vehicle around the full loop and steering no longer saturates. Only then should supervised actuation resume.

Closed-track validation sequence

Step	Measurement	Decision enabled
1	Verify LiDAR, IMU, and map-to-base_link	Confirm localization readiness
2	Save same-session PBStream and PGM/YAML	Lock coordinate origin
3	Generate track-specific 500-point centerline	Create compatible waypoints
4	Run Pure Pursuit on debug command topic	Check targets and steering
5	Supervised low-speed vehicle test	Complete real-car validation